

Akreti Sharma

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EDUCATION

University of Waterloo

Bachelor of Computer Science (Honours), Co-op

Waterloo, ON

Sep. 2026 – Present

St. Augustine Secondary School

Ontario Secondary School Diploma

Brampton, ON

Sep. 2022 – Jun. 2026

EXPERIENCE

Programming Research Intern

Jul. 2026 – Aug. 2026

Sunnybrook Research Institute – Focused Ultrasound High School Summer Research Program

Toronto, ON

- Developed Python-based software tools to automate coordinate generation and visualization for an automated fluid dispensing system used in focused ultrasound research.
- Implemented coordinate generation and file output workflows for programmable dispensing patterns, enabling precise placement of epoxy elements and supporting automated fabrication.
- Developed Python scripts to interface with Radiation Force Balance (RFB) equipment, collect experimental measurements, process sensor data, and calculate acoustic power output.
- Built data visualization workflows using NumPy and Matplotlib to analyze measurement stability, baseline drift, and experimental results.

Assistant Teacher

Sep. 2025 – Present

Spirit of Math

Brampton, ON

- Provide individualized math support for students in grades K–11 while assisting teachers with classroom organization and management.
- Track student progress and provide constructive feedback to support learning outcomes.
- Assist with organizing activities and supporting daily classroom operations.
- Communicate effectively with students, teachers, and parents to support a positive learning environment.

LEADERSHIP & EXTRACURRICULARS

HOSA Canada

Dec. 2022 – Present

Chapter President, Education Lead & Competitor

- Led chapter initiatives and coordinated activities for 70+ members.
- Organized meetings, educational workshops, and fundraising initiatives.

Coding Club

Sep. 2025 – Present

President

- Led weekly meetings teaching programming concepts and guiding collaborative coding projects.
- Planned out beginner-friendly workshops focused on developing foundational coding and problem-solving skills.

PROJECTS

Needy – Hack the North 2026 | AI Inventory & Procurement

Sep. 2026

- Built an AI inventory assistant using React, FastAPI, OMNI, and a Luxonis OAK camera to identify inventory, compare prices across four retailers, and create Zip purchase requests with automated vendor selection.

Goodminton – Ignition Hacks | Winner – Best Use of Roboflow

Aug. 2025

- Developed an AI-powered badminton match analyzer using YOLO-based motion tracking and built a real-time scoreboard with Next.js and Tailwind CSS.

EchoLens – Design Like a Girl Hackathon | Assistive Tech Prototype Winner

Jul. 2025

- Led development of a wearable prototype that assists visually impaired users through proximity-based audio feedback.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Swift, SQL, HTML, CSS, Lua

Frameworks: Next.js, Node.js, TailwindCSS, YOLOv5, OpenCV, PyTorch

Developer Tools: Git, VS Code, Xcode, Looker Studio

Data Science: NumPy, Matplotlib, Data Processing, Machine Learning, Computer Vision